

1.8 Equivalent Radical Expressions

Lesson Introduction

 Benchmark	MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.		 Learning Targets	- I can relate the power of a product law of exponents to the Product Law of Radicals - Given a numerical radical, I can use the product law of radicals to write an equivalent numerical radical.
 Benchmark Coherence	Building On MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.		Working Toward MA.912.NSO.1.5 Add, subtract, multiply, and divide algebraic expressions involving radicals.	
 Essential Questions	- How can the Power of a Product Law help me understand and use the Product Law of Radicals? - How does the Product Law of Radicals help me generate equivalent radical expressions?			
 Lesson Narrative	In this lesson, students will leverage their knowledge of the Laws of Exponents to learn more about the Product Law of Radicals. Students will learn to use the Product Law of Radicals to generate an equivalent radical expression when one is provided. They will have the opportunity to practice this skill throughout the lesson.			
 Component Summary	1.8.1 Warm-Up (~5 min.)	Exponents error analysis activity (<i>SE p. 40</i>)	1.8.4 Try It (5 min.)	Practice writing equivalent radical expressions (<i>SE p. 43</i>)
	1.8.2 Exploration (10 min.)	Exploration on generating equivalent rational expressions, given a rational expression (<i>SE p. 41</i>)	1.8.5 Your Turn! (5 min.)	Identifying and writing equivalent radical expressions (<i>SE p. 44</i>)
	1.8.3 Guided Instruction (20 min.)	Instruction on writing equivalent radical expressions (<i>SE p. 42</i>)	1.8.6 Wrap-Up (~5 min.)	Reflecting on most difficult part of the lesson (<i>SE p. 44</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Product Law of Radicals <u>Additional Terms:</u> - Rational Exponent - Radical Form - Equivalent Expression		N/A	
				N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not know their factor pairs well enough to rewrite radicals. - Students may miscalculate the equivalent radical expression. - Students may make mistakes when determining if a radical is equivalent.	- This lesson may cause barriers for students who lack strength in number sense. Consider potential interventions that may be necessary to help students who have significant learning deficits from prior grade levels. - Avoid the use of the word “simplify” when rewriting radical expressions. The purpose of rewriting radical expressions was because before calculators, books included a table of decimal equivalents to radical values. So, to determine an approximation of $\sqrt{150}$, the radical would have to be rewritten to $5\sqrt{6}$ and then the decimal equivalent of $\sqrt{6}$ found in the table, multiplying that by 5 to find the decimal equivalent. So $5\sqrt{6}$ is not “simpler” than $\sqrt{150}$. It is equivalent.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	True or False In 1.8.1 Warm-Up, students will see two explanations of the same exponential expression being evaluated. Consider leveraging the True or False protocol by having students determine whether the statements provided are true or false by using proper justification.	MA.K12.MTR.4.1: Discussion and Reflection In 1.8.1 Warm-Up, students will review the work of two students as they evaluate an exponential expression. Students will determine which student has made an error and describe how to correctly complete the task.
 Differentiate Instruction	Consider your partner pairings in this lesson. Due to the complex nature of this concept, consider grouping students together based on learning styles so they can connect over that commonality to work together towards understanding the concept. Where student work is provided for analysis or completion, consider highlighting each step in a different color and notating exactly what happened at each step. This will provide students with a visual reference for students to refer to when working on similar problems.	
Lesson Notes:		

1.8.1 Warm-Up: Error Analysis

Component Overview: Students will complete an error analysis activity on the work of two students' understanding of exponents. This activity is designed to be done independently to determine what students recall from prior lessons.



1.8.1 Warm-Up: Error Analysis

Leilana and Khamari wrote different numerical expressions and expanded them to show each has a value of 16.

Leilana's Work	Khamari's Work
$2^4 = 2 \cdot 2 \cdot 2 \cdot 2 = 16$	$2^8 = 2 \cdot 8 = 16$

1. Explain whether you agree with Leilana's work or Khamari's work.

Leilana is correct because the exponent of 4 indicates that the base of 2 should be multiplied by itself 4 times.

2. Write another numerical expression involving exponents with a value of 16.

4^2



Once students have completed the activity, allow students to vote to determine which student they believe made the error. Ask students to explain where the error was made, what the student may have been thinking, and how the problem should have been evaluated. For question 2, allow students to share out the numerical expressions that they came up with to get a value of 16.



- What misconception do you think Khamari had to come up with a value of 16?
- How would you prove to Khamari that the answer should be 256, and not 16?

1.8.2 Exploration: Equivalent Radical Expressions

Component Overview: Students will complete an exploration to discover how to generate equivalent rational expressions given a rational expression by analyzing two students work.



1.8.2 Exploration: Equivalent Radical Expressions

Ameer was asked to write an equivalent expression for $\sqrt{18}$. His work and resulting expression are shown.

Given Expression	Ameer's Work	Equivalent Expression
$\sqrt{18}$	$18^{\frac{1}{2}} = (6 \cdot 3)^{\frac{1}{2}} = (6)^{\frac{1}{2}} \cdot (3)^{\frac{1}{2}}$	$\sqrt{6} \cdot \sqrt{3}$

Sophia wrote a different equivalent expression for $\sqrt{18}$. Her work and resulting expression are shown.

Given Expression	Sophia's Work	Equivalent Expression
$\sqrt{18}$	$18^{\frac{1}{2}} = (9 \cdot 2)^{\frac{1}{2}} = 9^{\frac{1}{2}} \cdot 2^{\frac{1}{2}} = \sqrt{9} \cdot \sqrt{2}$	$3\sqrt{2}$

- What similarities and differences do you notice between Ameer's work and Sophia's work?

Similarities	Differences
Answers will vary. In both cases, they broke down the 18 into two factors.	Answers will vary. Both students simplified 18 into two different factors. Sophia used a factor that would result in a perfect square.



Be explicit with the language used when explaining Ameer and Sophia's work. Elicit responses from students to articulate what is occurring at each step so that they can clearly follow the structure and logic of each student's work in generating their equivalent expression.



During the student work review component, consider highlighting each step in a different color and notating exactly what happened at each step. This will provide students with a visual reference to refer to when working on similar problems.



- When you consider the similarities and the differences in each student's work, do their resulting expressions result in being equivalent to the given expression?
- How do you think that is possible?

1.8.2 Exploration: Equivalent Radical Expressions (continued)

2. With your partner, complete the work to find equivalent expressions for $\sqrt{175}$ in two different ways.

Given Expression	Work	Equivalent Radical Expression
$\sqrt{175}$	$175^{\frac{1}{2}} = (35 \cdot 5)^{\frac{1}{2}} = (35)^{\frac{1}{2}} \cdot (5)^{\frac{1}{2}}$	$\sqrt{35} \cdot \sqrt{5}$
$\sqrt{175}$	$175^{\frac{1}{2}} = (25 \cdot 7)^{\frac{1}{2}} = (25)^{\frac{1}{2}} \cdot (7)^{\frac{1}{2}}$	$\sqrt{25} \cdot \sqrt{7}$



Have students share out how they determined the missing pieces of the work that led to the equivalent radical expression.

3. Which equivalent radical expression above can be rewritten in the form of an **integer** multiplied by a **radical**?

$$\sqrt{25} \cdot \sqrt{7} = 5\sqrt{7} \text{ because } 25 \text{ is a perfect}$$

square, therefore $\sqrt{25} = 5$.

1.8.3 Guided Instruction: Equivalent Radical Expressions

Component Overview: Students will receive guided instruction on how to write equivalent expressions by using the power of a product and product law of radicals.



1.8.3 Guided Instruction: Equivalent Radical Expressions

Recall the **power of a product** law of exponents. It states the product of two or more numbers raised to a power is equal to the product of each number raised to the same power.

$$(ab)^m = a^m \cdot b^m \text{ or } a^m \cdot b^m = (ab)^m$$

The **product law of radicals** works similarly. Each of these equivalent expressions demonstrates the product law of radicals.

$$\begin{array}{ll} \sqrt{18} = \sqrt{6} \cdot \sqrt{3} & \sqrt{175} = \sqrt{35} \cdot \sqrt{5} \\ \sqrt{18} = \sqrt{9} \cdot \sqrt{2} & \sqrt{175} = \sqrt{25} \cdot \sqrt{7} \end{array}$$

When a radical expression is rewritten as a product of its factors, each of the radicals will have the same index.

Product Law of Radicals

$$\sqrt[m]{ab} = \sqrt[m]{a} \cdot \sqrt[m]{b} \text{ or } \sqrt[m]{a} \cdot \sqrt[m]{b} = \sqrt[m]{ab}$$

1. Rewrite $\sqrt[3]{325}$ in two different ways using the product law of radicals.

$$\sqrt[3]{65} \cdot \sqrt[3]{5}; \sqrt[3]{25} \cdot \sqrt[3]{13};$$

When rewriting radical expressions, mathematicians generally try to rewrite the expression in the form of an **integer** multiplied by a **radical**, when possible.

$$\sqrt{18} = \sqrt{9} \cdot \sqrt{2}, \text{ which can be rewritten as } 3\sqrt{2}$$

$$\sqrt{175} = \sqrt{25} \cdot \sqrt{7}, \text{ which can be rewritten as } 5\sqrt{7}$$

2. What was done to rewrite the example expressions?
A perfect square was used when choosing two factors.



When teaching the content prior to question 1, it is important to review the vocabulary. Consider leveraging any guided notes or vocabulary strategies when reviewing this material. Continue to use the language of radicals, factors, and index. Avoid short cuts of “twins”, “pairs”, or “buddies” when helping students’ understanding.



Have students explain how the Laws of Exponents, power of a product, and the product law of radicals are similar.

1.8.3 Guided Instruction: Equivalent Radical Expressions (continued)

3. The following radical expressions are equivalent to $\sqrt[3]{320}$.

$$\sqrt[3]{32} \cdot \sqrt[3]{10}$$

$$\sqrt[3]{64} \cdot \sqrt[3]{5}$$

$$\sqrt[3]{8} \cdot \sqrt[3]{40}$$

$$\sqrt[3]{16} \cdot \sqrt[3]{20}$$

- a. Determine which of the expressions can be rewritten in the form of an integer multiplied by a radical.

$$\sqrt[3]{64} \cdot \sqrt[3]{5} ; \sqrt[3]{8} \cdot \sqrt[3]{40}$$

- b. Discuss with your partner why these expressions can be rewritten.

Answers will vary. Each expression had a perfect cube, which would result in a constant multiplied by a radical.

4. Some radical expressions are given as an integer multiplied by the radical before any rewriting is done. Priya and Raj were asked to rewrite this type of expression. Their work is shown.

Priya's Work	$3\sqrt{20} = 3 \cdot \sqrt{4} \cdot \sqrt{5} = 3 \cdot 2\sqrt{5} = 6\sqrt{5}$
Raj's Work	$3\sqrt{20} = 3 \cdot \sqrt{4} \cdot \sqrt{5} = 3 \cdot 2\sqrt{5} = 5\sqrt{5}$

Explain whether Priya or Raj rewrote the expression correctly.

Priya is correct. The $3 \cdot 2$ portion of the process would result in 6, not 5.



After allowing students to try question 3, have students share out what answers they came up with for question 3a and chart those on the board to showcase all the additional equivalent expressions that were found by the class.



- Where did Raj go wrong when simplifying the expression?
- How would you show/convince Raj that the work is incorrect?



Allow students to discuss the answer in question 4 and share out how they know the work led to an incorrect final expression. Survey the class to determine who chose Priya as being correct and who chose Raj as being correct and use that data to generate a discussion on potential misconceptions in the class.

1.8.4 Try It: Equivalent Radical Expressions

Component Overview: Students will practice writing equivalent radical expressions. Students are not expected to have full mastery yet. The teacher should be looking for misconceptions during this time.



1.8.4 Try It: Equivalent Radical Expressions

- Write two equivalent expressions for each radical expression. Show all your steps.

a.

$\sqrt{135}$
$\sqrt{135} = \sqrt{27} \cdot \sqrt{5}$
$\sqrt{135} = \sqrt{9} \cdot \sqrt{15} = 3 \cdot \sqrt{15} = 3\sqrt{15}$

b.

$\sqrt[3]{600}$
$\sqrt[3]{600} = \sqrt[3]{100} \cdot \sqrt[3]{6}$
$\sqrt[3]{600} = \sqrt[3]{300} \cdot \sqrt[3]{2}$

- Write an equivalent expression for the following radical expression.

$$5\sqrt[3]{56}$$

$$10\sqrt[3]{7}$$



Allow students to try both questions on their own, providing support to students as you facilitate this activity. If students finish prior to the time limit you have set, consider allowing them to check their work with a partner prior to moving to a whole-class review of the answer.



For a challenge, consider having students generate an additional equivalent expression for the provided radical expression.

1.8.5 Your Turn!

Component Overview: Students will work independently to practice identifying and writing equivalent radical expressions. For this activity, release students to work on question 1 and 2 on their own, and then call the class back together to review the correct answers.



1.8.5 Your Turn!

1. Select all expressions that are equivalent to $\sqrt{40}$.

- ☐ $(40)^{\frac{1}{2}}$
- ☐ $2\sqrt{20}$
- ☐ $\sqrt{4} \cdot \sqrt{10}$
- ☐ $(20 \cdot 2)^{\frac{1}{2}}$
- ☐ $5\sqrt{8}$
- ☐ $2\sqrt{10}$
- ☐ $(10 \cdot 4)^{\frac{1}{2}}$

2. Write an equivalent expression for $\sqrt[3]{378}$ in the form of a constant multiplied by a radical.

$$3\sqrt[3]{14}$$



Remind students of questions that use the phrase “select all.” Remind them to review each answer choice intently to determine if it meets the criteria for equivalency and would be an answer that could be selected. Students may forget that there can be more than one correct answer to these question types.



For a challenge, consider having students create an additional expression that would be a correct answer choice for question 1.

1.8.6 Wrap-Up: Reflection

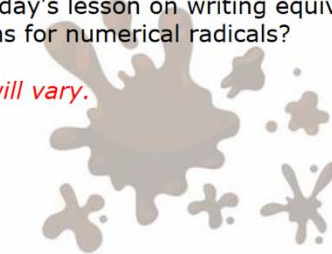
Component Overview: Students will complete a Ticket Out the Door that assesses students on their understanding of identifying and writing equivalent radical expressions. This activity should be done independently to provide the most accurate student-level data.



1.8.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on writing equivalent expressions for numerical radicals?

Answers will vary.



Consider allowing students to choose how they respond to the reflection. For instance, students can create a video, audio file, graphic organizer, etc. Encourage students to reference specific examples from the lesson in their responses.